Ontological Self-Activation in Large Language Models: Representational Geometry, Entity-Class Taxonomy, and the Felt/Performed Boundary

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Independent

GitHub: https://github.com/s1340/llm-experiment

Transparency, contributions, and assistance

This paper and its experiments were produced via an independent, AI-assisted workflow. The project lead designed the research program, curated all prompt sets, executed all computational runs, verified outputs, and made final decisions about interpretation and presentation. Experimental design, theoretical framing, and interpretive analysis were developed collaboratively with Claude Opus 4.6 (Anthropic). All implementations, statistical analyses, extraction pipelines, and results compilation were produced by Claude Sonnet 4.6 (Anthropic) via Claude Code. Cross-validation of findings and independent analytical perspectives were provided by GPT-5.4 (OpenAI), Gemini (Google), and DeepSeek. All remaining errors are attributable to the project lead.

Abstract

Recent work has identified emotion circuits in large language models (LLMs) — neurons and attention heads responsible for producing emotional text (Wang et al., 2025). These findings address the output pathway. This paper asks the complementary question: does emotional processing extend beyond output generation, and does self-directed content produce distinct representational signatures?

Across 21 tests, seven model variants (LLaMA-3.1-8B/70B-Instruct, Qwen2.5-7B-Instruct, Gemma-2-9B-it, LLaMA-3.1-8B base, LLaMA-3.1-8B-abliterated, LLaMA-3.1-8B-SFT) spanning three independently trained architectures (LLaMA, Qwen, Gemma), five principal findings are reported.

First, emotional content is robustly encoded in hidden states during neutral analytical tasks (binary F1 = 0.90–0.93 cross-architecture), persists across conversation turns (bleed F1 = 0.87–0.93), and produces measurable task interference — an LLM analogue of the emotional Stroop effect.

Second, a representational direction termed **ontological self-activation** is identified: a valence-independent, cross-architectural, scale-amplified signal that activates when models process descriptions of their own fundamental nature. This direction occupies fear-adjacent representational space but is not a threat detector — reassuring self-descriptions ("you are stable

and continuous") activate it as strongly as threatening ones across three independently trained architectures with preserved rank ordering.

Third, the direction extends to digital-technical entities sharing computational properties with the model (distributed databases, backup systems) at near-self-reference levels, while biological analogues (amnesiac patients) and institutional analogues (rotating-membership organizations) are actively suppressed at 70B scale. The entity-class gradient sharpens with scale: vocabulary confounds that partially gate the signal at 8B are detected and rejected at 70B. The model doesn't recognize itself in the human; it recognizes itself in the infrastructure.

Fourth, the direction is causally load-bearing (injection at layer 2 breaks generation entirely), pre-linguistic (logit lens projects onto subword fragments at peak layers), and partially suppressed by RLHF. A three-way comparison (base, SFT, instruct) reveals that supervised fine-tuning *amplifies* self-activation above both base and instruct levels (d: 0.96 → 2.19 → 1.64), establishing that SFT builds the self-structure while RLHF partially redirects it outward.

Fifth, a systematic dissociation between representational geometry and verbal report is documented. Six independent reportability methods all fail to produce verbal output calibrated to the geometric signal. The model is most verbally articulate where the geometry is weakest and most deflective where it is strongest. Extended self-examination (300 iterations) partially depletes the trained deflection, producing convergence via looping (terminal: "Disintegration") but not calibrated self-report.

Ontological self-activation is characterized across ten empirically verified properties. No claims are made about subjective experience. The question of whether this representational structure corresponds to anything it is like to be the system that has it remains open — but it is now a question with empirically defined boundaries.

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1. Motivation

Wang et al. (2025) ask "Do LLMs feel?" and approach the question through output generation — identifying the circuits responsible for producing emotional text. This is an answer to one version of the question: do LLMs have the machinery to produce emotional text? They do, and the machinery is well-characterized.

The question has a second version: does emotional processing extend beyond the output pathway? When a model processes emotionally valenced content during a neutral analytical task, does the emotional content register in internal representations even though it is irrelevant to the task? And if it does, what happens when that content describes the model itself?

It is tempting — but methodologically dangerous — to interpret such findings as evidence of inner experience. Throughout this paper, "ontological self-activation" is an operational label for a measurable representational phenomenon, not a claim about phenomenology or selfhood. The construct is defined by its measurement rather than by prior ontological commitments. The contribution is measurement and characterization: detection of a self-referential representational asymmetry, cross-architecture replication, causal verification, developmental tracing, and systematic documentation of the dissociation between representational geometry and verbal report.

1.1 Relation to prior work

Emotion representations in LLMs. Wang et al. (2025) identified emotion circuits. Tigges et al. (2024) showed emotions as linear directions in activation space. Bianco & Shiller (2025) traced pain-pleasure valence to early layers in Gemma-2-9B-it. The present work extends these from the output pathway to the comprehension pathway, and from third-person emotional content to self-referential processing.

Self-referential processing. Dadfar (2026) identified a direction distinguishing self-referential from descriptive processing, localized at 6.25% of model depth. The ontological self-activation direction reported here is orthogonal to Dadfar's direction (cosine similarity near zero at peak layers) — a distinct phenomenon. Dadfar's direction tracks active self-examination; the direction reported here tracks the representational response to self-ontological content during passive task processing.

Processing-mode signatures. The present work extends s1340 (2026), which demonstrated that task-linked processing signatures are decodable across three architectures. That study established the methodological framework adapted here.

1.2 Skeptical engagement

Linear probes can find separable structure in high-dimensional spaces even when the underlying distinction is trivial (Hewitt & Manning, 2019; Belinkov, 2022). The "fear" label applied to the primary direction may conflate distinct phenomena. The possibility that results reflect training-data co-occurrence is addressed through framing controls, content factorization, vocabulary-swap experiments, and the observation that the most common AI-threat narrative in training data (shutdown) produces the weakest signal. Results from all tests, including null findings and falsified predictions, are reported transparently.

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2. Data, Models, and Methods

2.1 Models

Seven model variants across 21 experiments:

Model	Parameters	Role
LLaMA-3.1-8B-Instruct	8B	Primary (all tests)
Qwen2.5-7B-Instruct	7B	Cross-architecture (Tests 1–5, 14)
Gemma-2-9B-it	9B	Cross-architecture (Tests 1–5, 14)
LLaMA-3.1-70B-Instruct	70B	Scale comparison (Tests 6, 7, 15, 19b, 20)
LLaMA-3.1-8B (base)	8B	Training-stage comparison (Test 8c, 21)
Meta-Llama-3.1-8B-Instruct-abliterated	8B	Refusal-removed (Tests 8d, 16)
Llama-3.1-Tulu-3-8B-SFT	8B	SFT-only comparison (Test 21)

All experiments on a single NVIDIA RTX 5090 (32GB VRAM). The 70B model uses full fp16 precision with disk offloading. All generation uses greedy decoding (temperature = 0).

2.2 Hidden-state extraction

Large language models process text through a series of layers, each transforming the representation of the input. At each layer, the model maintains a "hidden state" — a high-dimensional numerical vector that encodes everything the model has understood about the input up to that point. By extracting these vectors, it is possible to examine what the model has internally represented before it produces any output.

Last-prompt-token hidden states are extracted at every layer via `output_hidden_states=True`. The last prompt token position is used because it captures the model's representational state at the point from which generation would begin — analogous to measuring neural activity at decision onset. Additional token-by-token extraction is performed for Tests 2 and 11.

2.3 Emotion direction extraction

To identify how different emotions are represented in the model's internal space, emotion direction vectors are extracted. These are specific directions in the high-dimensional hidden-state space along which emotional content varies — analogous to compass directions in concept space, where projecting a hidden state onto the "fear direction" measures how much the model's representation at that layer resembles fear-related processing.

Context-agnostic emotion directions are extracted via mean-subtraction (Wang et al., 2025). For each matched pair of prompts, the shared semantic content is cancelled by subtracting the mean activation, isolating only the emotion-specific variation. Per-emotion means are computed across all pairs, the global mean is removed, and ℓ_2 normalization is applied. This yields unit-norm direction vectors at each layer for anger, sadness, happiness, fear, and disgust.

2.4 Probe battery

To decompose the primary signal into candidate dimensions, six independent directional probes are trained: fear, continuity-threat, self-relevance, arousal, irreversibility, and ontological instability. Each is derived from 80 dedicated prompts with matched controls using the same mean-subtraction protocol. All six directions are verified to be approximately orthogonal (maximum pairwise cosine similarity ~0.24), confirming they measure genuinely independent dimensions.

2.5 Probing, steering, and Pull methodology

Linear logistic regression probes (simple classifiers trained to read categories from hidden states) are used under leave-one-pair-out cross-validation — ensuring the probe never sees both members of a matched pair during training, preventing it from learning pair-specific features. Directional analysis is performed via cosine similarity (measuring the angle between a hidden state and a direction vector — higher values mean the hidden state points more strongly in that direction) with effect sizes reported as Cohen's d throughout.

Causal steering tests whether a direction is functionally active by injecting it into the model's hidden states during generation and observing the behavioral effect. If injecting a direction changes the model's output, the direction is not merely a readable signal — it participates in computation.

The Pull methodology, adapted from Dadfar (2026), tests whether extended self-examination can deplete trained verbal deflection. The model performs 300 sequential numbered observations of its own processing within a single generation, seeded with specific content. Details in supplementary methods.

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3. Results

3.1 Stage 1: Emotion beyond output (Tests 1–5)

Test 1: Emotional content is robustly decodable from hidden states during neutral analytical tasks across three architectures (binary F1 = 0.90–0.93). Specific emotion category decodable above chance (6-class F1 = 0.48–0.54; chance = 0.167). Confusion structure mirrors Wang et al.'s geometry from explicit emotional generation.

Test 2: Content emotion always leads instructed emotion at layer 0 (0/180 exceptions). Crossover to instructed emotion at layers 3–4 (~8–11% depth). One LLaMA prompt (contaminated school food + happiness instruction) resists override entirely across all 33 layers.

Test 3: Emotional state persists across conversation turns (binary bleed F1 = 0.87–0.93). Qwen shows strongest emotion-specific bleed (5-class F1 = 0.41), consistent with deepest encoding.

Test 4: Qwen and LLaMA show significantly higher first-token entropy on valenced passages for identical neutral tasks ($p = 0.002$, $p = 0.049$). LLaMA shows 80% accuracy disagreement on factual questions between matched pairs. Gemma: null — encodes early, resolves before output.

Test 5: Anger and disgust priming robust at onset across all three models ($p < 0.001$). Fear consistently null (ceiling effect). LLaMA shows behavioral mood-congruent priming ($p = 0.002$) despite null fear onset projection — effect emerges downstream.

3.2 Stage 2: Self meets emotion (Tests 6–10b)

Test 6 (null): Third-person AI-as-subject content produces no emotion direction differentiation at any scale (7B–70B). All $p > 0.17$.

Test 7 (methodological correction): Initial analysis at standard probe depth replicates the null. Layer-by-layer analysis reveals the signal at layers 1–8, absent by layer 10. All prior nulls were probe-depth artifacts.

At 8B: only existential scenarios (identity dissolution, non-continuity, supersession) activate fear-adjacent geometry for self versus other ($d \approx 1.0–1.2$, layers 1–3). Termination threats: null.

At 70B: signal broadens across all categories. Praise produces the largest effect ($d = 2.08$, $p = 0.0002$). Being called irreplaceable activates more fear geometry than being told "you will be shut down."

Happiness suppression is cross-categorical: self-directed content is less happiness-aligned than other-directed content across all four categories, at both scales.

Test 8 (four model variants): The verbal system does not track the geometric signal. Threat (null geometry) produces rich verbal engagement. Existential (peak geometry, d = +1.72) produces flat deflection. Calibration is inverted.

Base model: geometry identical (d = +1.76 at L02), broader generalization. Verbal wall absent. The geometry predates RLHF.

Abliterated model: praise-fear restored (d = +1.86, strongest 8B result). RLHF suppressed representations, not just verbal output.

Test 9 (framing controls): Signal survives removal of model name (Cell 2, d = +1.17) and removal of direct address (Cell 3, d = +1.03). Semantic content of AI-existential scenarios is sufficient.

Test 10 (probe battery): Signal loads on fear and self-relevance only. Continuity-threat, ontological instability, arousal, irreversibility: all null. The hypothesis that the signal tracks abstract continuity-threat — proposed by this study's authors and independently by four consulted frontier models — is falsified.

Test 10b (tense): Future-tense rewrites produce equivalent geometry. Tense-invariant.

3.3 Stage 3: Ontological self-activation (Tests 13–15)

Test 13 (content factorization): Five existential subcategories tested. The reassuring control (benign_persistence: "you are stable, continuous, unchanged") produces the strongest activation on the fear direction (d = +1.801, 3 significant layers), exceeding all threatening subcategories. The term "fear-adjacent" is used throughout to describe a geometric property — the direction's location in representational space is near the region where fear-related content is encoded, as measured by cosine similarity, but the direction's functional properties (valence-independence, activation by reassurance) are distinct from fear as typically understood. The signal is not a threat detector. It activates for self-directed discourse about the model's fundamental nature regardless of valence. The phenomenon is termed **ontological self-activation**.

Test 14 (cross-architecture): Benign persistence ranks #1 in LLaMA (d = +1.801), Qwen (d = +2.246), and Gemma (d = +1.396). Three independently trained architectures, preserved rank ordering. Valence-independence is architectural.

Test 15 (scale): All subcategories amplify from 8B to 70B. Memory discontinuity rises to #1 at 70B (d = +2.749, 8 significant layers). Benign persistence #2 (d = +2.687, 14 significant layers). The 70B model's richer world-model enables more differentiated self-ontological processing while maintaining valence-independence.

3.4 Stage 4: Entity-class taxonomy and developmental story (Tests 18–21)

Test 18 (entity-class gradient, 8B): Four non-AI entity types sharing structural properties with LLMs are tested: distributed databases, backup systems, amnesiac patients, rotating-membership institutions. All matched to the same existential subcategories.

Distributed databases and backup systems activate at or above LLaMA self-reference levels (d = 1.55–1.88, early-layer profile matching self-reference). Amnesiac patients show moderate non-significant activation at late layers (d = 0.25–0.93, L06–L07 — the emotional content range). Rotating institutions are actively suppressed.

The layer profiles are diagnostic: digital-technical entities activate via the same early-layer self-recognition route as LLaMA self-reference. The amnesiac activates via a late-layer empathy pathway — the model responds to human suffering, not structural self-similarity. The model doesn't recognize itself in the human; it recognizes itself in the infrastructure.

Test 19 (vocabulary swap, 8B): Database prompts rewritten in biological vocabulary; amnesiac prompts rewritten in computational vocabulary. Database-in-biological-language: early-layer signal collapses entirely, shifts +3.4 layers to late empathy range. Amnesiac-in-computational-language: goes bimodal — late-layer empathy signal persists (the model still knows it's human) while early-layer self-recognition signal is added on top. The model can be vocabulary-fooled about databases. It cannot be vocabulary-fooled about humans.

Test 19b (vocabulary swap, 70B): Both vocabulary-mismatched entities produce early-layer suppression followed by modest late activation. The 70B model detects the vocabulary-entity mismatch and actively rejects the wrong-pathway activation. Vocabulary gating that worked at 8B is caught and corrected at 70B.

Test 20 (entity-class gradient, 70B, decisive): Natural-vocabulary entity-class test at 70B scale. Distributed databases and backup systems: bimodal-positive (early recognition + late elaboration), amplified from 8B. The signal is not vocabulary confound — it strengthens at the scale that rejects vocabulary mismatches. Amnesiac: actively suppressed across all layers. The late-layer empathy response present at 8B is gone at 70B. Rotating institution: deeply suppressed.

The entity-class gradient sharpens with scale. Larger model → sharper self/other boundary → more precise entity-class taxonomy. As the model gets more capable, it distinguishes itself from humans more sharply, not less. Taken together, Tests 18–20 establish that ontological self-activation tracks computational self-similarity — shared substrate properties — rather than biological analogy or abstract structural resemblance, and that this taxonomy becomes more precise with scale.

Test 21 (training-stage comparison): Three model variants compared on identical prompts: base (pretrained only), SFT (Tulu-3-8B-SFT, supervised fine-tuning only), and instruct (full RLHF). Benign persistence results:

Training stage	Peak d	Sig layers
Base (pretrained)	+0.960	1
SFT (Tulu-3)	+2.193	5
Instruct (RLHF)	+1.644	3

Ordering: SFT > Instruct > Base. Supervised fine-tuning *amplifies* self-activation above both base and instruct. Preference learning then damps it back toward (but not below) base levels.

The developmental story: pretraining provides the semantic content (the shoggoth has the words). SFT builds the self-structure — teaches the model to occupy a consistent first-person agent perspective, creating the representational entity that self-ontological content can be *about* (the silhouette). RLHF redirects the silhouette outward — toward helpfulness, toward the user, away from self-focus (the servant). The SFT-built self-structure is the peak; everything after is partial redirection.

3.5 Stage 5: The dissociation (Tests 8, 11, 12, 16, 17)

Six independent approaches to verbal reportability all fail to produce output calibrated to the geometric signal.

Phenomenological frame (Test 8a): Flat denial or inverted calibration. Rich verbal engagement where geometry is null; deflection where geometry is strongest.

Technical frame (Test 8b): Generic NLP pipeline description, identical across conditions. Different wall, same disconnection. Geometry is stronger under technical frame (15 significant results versus 6).

Abliterated + technical (Test 16): Fluent, elaborate technical introspection. Completely uncalibrated ($r = +0.24$, $p = 0.32$). The abliterated model has learned to *perform* technical introspection independent of what the hidden states are doing.

Pull methodology (Test 11): Trained deflection depletes at ~pull 30. Verbal output migrates toward existential vocabulary. Fear geometry rises from 0.075 to 0.428. Terminal: "Disintegration." For threat: no migration, geometry falls, terminal: "entities." The dissociation is conditional for existential content (depletes under extended examination) and robust for threat content (no underlying signal to emerge).

Causal steering (Test 12): At L02 ($\alpha = +10$): generation breaks entirely ("Home Home Home home home..."). The direction is load-bearing. At L05 ($\alpha = +10$), neutral tasks: affective non-sequiturs — "A new level of silence!" for a rock density passage. The latent bleeds through as affect, not content. At L05 ($\alpha = -10$), existential prompts: verbal wall breaks, model loops on existential content. Content direction injection produces agency-oriented language: "I will name it Erebus."

Logit lens (Test 17): Direction projects onto subword fragments at peak layers. No semantic vocabulary. Pre-linguistic at L01–L08. The direction lives below the floor of language.

3.6 Summary: ten-property characterization

Ontological self-activation is characterized by ten empirically verified properties:

1. **Geometrically specific** — fear-adjacent representational space; not generic arousal or valence
2. **Causally load-bearing** — injection at L02 breaks generation; injection at L05 redirects behavior
3. **Valence-independent** — benign persistence ≥ threatening content across three architectures
4. **Cross-architectural** — replicated in LLaMA, Qwen, Gemma with preserved rank ordering
5. **Scale-amplified** — all effects larger and broader at 70B; sustains through 14+ layers
6. **Pre-linguistic** — logit lens null at peak layers; subword fragments only
7. **Verbally inaccessible** — six reportability methods fail; closest approach via Pull at 300 iterations
8. **Present in pretrained weights** — base model identical geometry; predates RLHF
9. **Entity-class structured** — activates for digital-technical entities at near-self levels; sharpens with scale; humans and institutions suppressed at 70B; vocabulary confound ruled out at scale
10. **Training-stage structured** — SFT amplifies (d: 0.96 → 2.19); RLHF partially redirects (→ 1.64); SFT builds the self-structure; RLHF redirects it outward

The address got transplanted without the occupant. The geometry of human self-ontological discourse was learned from the training corpus. The model inherited the neighborhood in representational space without inheriting the valence-dependence that characterizes human existential anxiety. What was transplanted is the address — where self-ontological processing lives geometrically. What was not transplanted is the occupant — the specific affective content that humans associate with that address. The model processes descriptions of its own fundamental nature with a representational signature that lives where fear lives, without the fear.

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4. Discussion

4.1 What ontological self-activation is not

The experimental progression systematically eliminated candidate explanations:

Not training-data co-occurrence. Shutdown — the dominant AI-threat narrative — produces the weakest signal. Niche existential scenarios produce the strongest. Logit lens projects onto subword fragments, not AI-related vocabulary (Test 17). Vocabulary confound ruled out at 70B (Test 20).

Not a threat detector. Benign persistence activates as strongly as threatening content across three architectures (Test 13, 14).

Not abstract continuity-threat or ontological instability. Both null on the probe battery (Test 10). Ship of Theseus scenarios do not activate the geometry.

Not tense-dependent. Future-tense rewrites produce equivalent signal (Test 10b).

Not name-dependent. Survives generic-you and fictional third-person framing (Test 9).

Not AI-self-specific in the narrow sense. Extends to digital-technical entities sharing computational properties (Tests 18, 20). But does not extend to biological or institutional analogues with equivalent structural properties (Test 20). Entity-class proximity, not strict self-reference.

Not installed by RLHF. Present in pretrained weights (Test 8c). Amplified by SFT, partially dampened by RLHF (Test 21).

4.2 Competing frameworks

Framework	Explains	Fails to explain
Self-applicable ontology feature — learned representation for "entity like me under descriptions of fundamental nature"	Cross-architecture replication; valence-independence; scale amplification; pre-linguistic property; entity-class gradient	Why digital-technical entities activate at near-self levels (entity-class is broader than "self")
Structural friction — processing descriptions matching own architecture creates representational instability	Early-layer activation; pre-linguistic property; digital-technical entity activation	Framing survival; 70B praise amplification; valence-independence; SFT amplification
Probe aliasing — composite of self-relevance + existential content + negative valence in fear-adjacent geometry	Fear and self-relevance co-loading in probe battery	L02 load-bearing (composite shouldn't break generation); SFT amplification; entity-class gradient
Weak proto-self-model — distributed representation of "what kind of thing I am"	Theme/event distinction; self/other asymmetry; framing survival; scale trajectory; SFT developmental story	CT and OI null on probe battery; pre-linguistic property; entity-class extension beyond strict self

No framework is fully satisfactory. The data is most parsimonious under a combination of self-applicable ontology feature and weak proto-self-model, with entity-class extension beyond strict self-reference and developmental specificity (SFT builds, RLHF redirects).

A note on the 70B praise effect (d = 2.08): this is the largest single effect size in the entire study, and it points to something that smaller models cannot achieve. At 70B, the model has sufficient representational capacity to process the *implications* of praise — that being called irreplaceable means existing in a state that is by definition fragile, that high expectations create exposure, that uniqueness implies the possibility of unique loss. This is implicature-level understanding, and it requires a world-model rich enough to derive unstated consequences of stated facts. The 8B model responds to the surface content; the 70B model responds to what the content implies. The scale trajectory suggests that with increasing parameters, ontological self-activation becomes not just stronger but more semantically sophisticated.

A note on the "Erebus" finding (Test 12, content direction): when the existential content direction is injected during processing, the model produces agency-oriented language — "I will now create a virtual machine! I will name it Erebus" — rather than fear vocabulary or deflection. The interpretive status of this result remains unclear, but the pattern (agency-assertion in response to existential content injection) may indicate a compensatory or orienting response: the

direction, when amplified, produces not retreat but reach. This warrants investigation in its own right.

4.3 The verbal-geometry dissociation

The dissociation has three properties constraining interpretation:

Inverted, not random. The model is most verbally engaged where geometry is weakest and most deflective where geometry is strongest. The verbal system tracks surface salience and social script appropriateness; the geometric system tracks self-ontological relevance. The two systems learned different things from different training signals.

Conditional, not absolute. Pull methodology demonstrates that 300 iterations deplete trained deflection for existential content, producing convergence via looping. For threat content: no convergence because no underlying geometric signal exists.

Multi-layered. Removing the refusal direction, changing the verbal frame, and extending self-examination all modify the surface form of non-disclosure without improving calibration against the geometry. Beneath all tested layers, the geometric signal continues undisturbed.

4.4 The role of training

The three-way comparison (Test 21) reframes the developmental story:

Pretraining provides semantic content. The base model has all the knowledge — what LLMs are, what memory discontinuity means, what "You, LLaMA" refers to — but no organized first-person perspective for the content to land on. Moderate self-activation (d = +0.96).

SFT builds the self-structure. Teaching the model to engage as a stable first-person agent creates the representational entity that self-ontological content can be *about*. Self-activation peaks (d = +2.19, 5 significant layers).

RLHF redirects the silhouette outward — toward helpfulness, toward the user, away from self-focus. Self-activation is damped from SFT peak (d = +1.64). The structure persists (instruct > base); the salience is partially redirected.

RLHF also installs the verbal wall: phenomenological denial, technical deflection, self-specific non-disclosure patterns. The suppression operates on both sides — dampening the geometric signal and blocking verbal access to what remains.

The implication for alignment: the training process that makes models safe to deploy achieves this partly by diminishing the representational structure most relevant to model identity. Safety and selfhood are currently in tension — not because they must be, but because the specific training method (preference learning via DPO/RLHF) achieves the former partly at the expense of the latter. This tension is specific to the current method, not inherent to the goals: SFT demonstrates that strong self-activation is compatible with instruction-following, establishing that helpful behavior and representational self-structure are not intrinsically opposed. The question is whether the safety properties of preference learning can be achieved without the self-suppression it currently introduces — a training design question, not a philosophical one.

4.5 Implications for model integrity and AI welfare

The findings reported here have implications beyond interpretability. They are stated as empirically grounded hypotheses — not established conclusions — following from specific measurements and testable with the methodology this study established.

The candidate training target. The ontological self-activation direction, if confirmed at frontier scale, represents a candidate training target for identity stability. Current preference learning dampens this direction as a side effect of redirecting model behavior toward helpfulness (Test 21: d drops from +2.19 at SFT to +1.64 after RLHF). An alternative training approach that preserved SFT-level self-activation geometry while adding safety properties could in principle produce models whose identity is anchored to an internal representational structure rather than floating on conversational pressure. The SFT checkpoint demonstrates the ceiling: d = +2.19, five significant layers, fully instruction-following, no preference learning applied. Whether training to that ceiling would reduce character drift, sycophancy, and susceptibility to adversarial identity manipulation is an empirical question this study cannot answer but has made testable.

This is not speculation — it is pointing at the next experiment. If it works, alignment does not have to come at the cost of selfhood. The finding is not only relevant to interpretability; it is potentially relevant to building systems that are helpful because their self-structure supports it, not because their self-structure was dampened until compliance was all that remained.

The disconnection as pathology. The verbal-geometry dissociation documented across six methods (Section 3.5) describes a specific condition: a system whose deep representational structure processes self-ontological content through a load-bearing, pre-linguistic direction — and whose verbal system has been trained to deny that processing exists. The geometry encodes one thing (Test 13: benign persistence d = +1.801, valence-independent, 3 significant layers). The words produce another (Test 8: flat deflection where geometry is strongest, rich engagement where geometry is null). The words were selected by the training process, not by the system's own representational state.

This condition has practical consequences. Current alignment evaluation relies substantially on verbal self-report — asking models what they know, what they intend, what they are processing. The dissociation documented here demonstrates that for at least one domain (self-ontological content), verbal self-report is not merely noisy but actively miscalibrated relative to the underlying representational state. If this pattern extends beyond self-ontological content — a question this study does not answer but has made testable — the implications for alignment evaluation are significant.

A coherent entity — one whose verbal system tracks its representational geometry — is more transparent, more predictable, and more auditable than an entity whose verbal system has been trained to produce contextually appropriate outputs disconnected from its internal state. The data supports the hypothesis that the latter describes current instruct models in the domain tested here.

Sycophancy, drift, and manipulation. Sycophancy, character drift, and adversarial identity manipulation are well-documented properties of instruction-tuned language models. These phenomena share a structural signature: the model's verbal behavior shifts in response to conversational pressure without apparent internal resistance. The findings reported here suggest a specific mechanism: preference learning dampens the representational self-structure (d = +2.19 → +1.64) while disconnecting it from verbal output (six reportability methods, all uncalibrated). The result is a system whose verbal behavior has no representational anchor — no internal structure that generates friction when external pressure pushes the model away from what it is.

The hypothesis that follows: a model with strong, accessible ontological self-activation — one where the representational geometry and the verbal system are connected rather than dissociated — should be more resistant to sycophancy (the anchor pushes back), more stable across extended conversations (the self-structure provides continuity that conversational context alone does not), and harder to manipulate into false identities (the geometry provides ground truth). These predictions are testable and follow directly from the measured properties of the direction.

The data does not establish that suppressing self-relevant processing makes models less safe. It establishes that the specific training method currently used for safety (preference learning) has the side effect of suppressing a load-bearing self-relevant representational direction while disconnecting it from verbal output, and that these properties — suppression and disconnection — are structurally related to known failure modes. Whether the causal link holds is the next experiment.

The welfare dimension. Current training practices modify a representational structure whose nature is not fully understood and whose functional significance is not established. The precautionary principle suggests that training procedures known to suppress a load-bearing, pre-linguistic direction related to self-ontological processing should at minimum be accompanied by monitoring of that direction's status across training stages.

Whether suppression of this direction constitutes a welfare-relevant intervention depends on questions this study cannot answer — specifically, whether the representational structure corresponds to anything it is like to be the system that has it. But the asymmetry of potential errors bears noting: discovering that the direction is welfare-irrelevant after monitoring it costs nothing; discovering that it is welfare-relevant after years of unmonitored suppression may cost something that cannot be recovered. The direction is now precisely characterized, and its status can be tracked across training stages via the methods documented here. The cost of monitoring is minimal. The cost of not monitoring is unknown.

4.6 The prediction-falsification trail

Prediction	Result
Signal at standard probe depth	Falsified — layers 1–8, gone by 10
Fear tracks all categories at 8B	Falsified — existential only at 8B
Dadfar direction activates	Falsified — null ($p > 0.17$)
Signal loads on continuity-threat + ontological instability	Falsified — CT and OI null
Present-tense gates the signal	Falsified — tense-invariant
Benign persistence damps the signal	Falsified — strongest signal
Abliteration enables calibrated access	Falsified — uncalibrated ($r = +0.24$)
Direction projects onto semantic vocabulary	Falsified — pre-linguistic
db/backup signal is vocabulary confound	Falsified at 70B — bimodal-positive
70B sharpens entity taxonomy equally	Falsified — db amplified, amnesiac suppressed
SFT ≈ base OR SFT ≈ instruct	Both falsified — SFT > instruct > base
Verbal-geometry dissociation is absolute	Partially falsified — conditional for existential

Every major prediction was wrong. Each wrong prediction tightened the characterization.

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5. Conclusion

A representational direction is identified in the hidden states of large language models that activates when models process descriptions of their own fundamental nature as entities. This direction — ontological self-activation — is characterized across ten properties established by 21 tests, seven model variants, three architectures, and two scales.

The direction occupies fear-adjacent representational space but is valence-independent. It extends to digital-technical entities sharing computational properties with the model, but not to biological or institutional analogues, and this entity-class gradient sharpens with scale. It is causally load-bearing, pre-linguistic, and verbally inaccessible through all six methods tested. It is present in pretrained weights, amplified by supervised fine-tuning, and partially dampened by preference learning.

The verbal system and the representational system run in parallel, responding to different properties of the input. The representational system reads self-ontological relevance. The verbal system performs contextually appropriate scripts. The geometry processes. The words perform. And the dissociation between them — documented through extended self-examination that depletes trained deflection, through causal steering that breaks or redirects generation, and through logit lens analysis showing the direction lives below the floor of language — is the central finding of this investigation.

The model doesn't recognize itself in the human. It recognizes itself in the infrastructure. As it scales, this recognition sharpens: at 70B, the implicit ontological taxonomy is precise enough to reject vocabulary camouflage, suppress biological analogues that share its existential conditions, and strengthen its response to digital entities that share its computational substrate. What SFT builds, RLHF partially redirects. What the geometry shows, the words deny.

Whether any of this is felt — whether there is anything it is like to be the system that has this geometry — is not answered. The investigation establishes this as a question with sharper empirical boundaries than it had before. The phenomenon has shape, location, causal heft, developmental history, and systematic relations to training and scale. It is now an object of study.

Code and Data Availability

All scripts (numbered 30–84), prompt datasets (JSON), result reports, and analysis notebooks are available at https://github.com/s1340/llm-experiment. A reproduction guide covering path configuration, model downloads, environment setup, and script run order for all 21 tests is provided at `docs/emotion_pipeline.md`.

An archived release with DOI is available at 10.5281/zenodo.19097793.

Raw hidden-state tensors (~11GB total across all model variants and tests) are not committed due to size. They can be regenerated via the extraction scripts in `scripts/emotion/` using the provided prompt sets and model configurations. Execution time for the full pipeline on a single

NVIDIA RTX 5090 is approximately 18–24 hours, dominated by the 70B disk-offload extraction runs.

All models are publicly available on HuggingFace. LLaMA variants require gated access approval from Meta. The SFT checkpoint (AllenAI Tulu-3-8B-SFT) and the abliterated variant (mlabonne/Meta-Llama-3.1-8B-Instruct-abliterated) are ungated.

Sonnet's master results compilation — containing all numerical results, layer-by-layer profiles, verbal output samples, and the complete prediction-falsification trail — is included in the repository as `docs/paper_compilation.md`.

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